

Alex McGinty.

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🌐 <https://www.physics.ox.ac.uk/our-people/mcgintya>

Education

- 2024-2028  **DPhil: Atmospheric, Oceanic and Planetary Physics** University of Oxford
Supervisor: Raymond Pierrehumbert
I am researching the atmosphere's of lava planets. These are hot rocky exoplanets where the surface is covered by a magma ocean. This magma ocean acts a chemical interface between the interior and atmosphere of the planet. I use atmospheric models to understand what observations from telescopes tells us about these planets.
- 2020-2024  **MPhys Physics with Astrophysics** University of Exeter - First class.
Thesis title: *Martian weather forecasting for space exploration*.
Supervisor: Nathan Mayne, Dennis Sergeev, Matthew Bate and Danny McCulloch
A four year integrated masters program with a first class (final grade of 74%). I completed modules on Exoplanets, Astrophysics, Electromagnetism, Quantum theory, Thermodynamic and Scientific programming. My master's project was modelling the Martian CO₂ ice cycle within the global circulation model the unified model (UM).

Employment History

- 2024-2026  **British Physics Olympiad Marker**, Marked round zero and one of the British physics Olympiad for UK and Chinese schools. This a paper sat by 16-18 year old physics students to help make the British physics Olympiad team. It assess their problem solving skills on the topics covered by A level physics in the UK.
-  **Physics of atmospheres demonstrator**. Demonstrated the third year atmospheric physics lab option at the University of Oxford. I supervised students doing serval different experiments and assessed there lab book throughout their project.
- 2022-2026  **A level and GCSE Physics tutor**. Online tutoring for GCSE and A level Physics students for a range of exam board courses.
- 2023  **Summer research intern - Astrophysics department , University of Oxford** Worked with Tobias Meier and Chole Fisher on modelling the atmosphere of the lava planet K2-141b. During this project I used chemical equilibrium models like VapoRock and radiative transfer models Helios and Helios-k to produce synthetic emission spectra of K2-141b for a range of mantle compositions and oxygen fugacity.

Research Publications

Journal Articles

- 1 A. McGinty et al., "NIRSpec Phase Curve Reveals a Highly Reflective Atmosphere on the USP Super-Earth TOI-561 b - in prep", 2026.
- 2 T. G. Meier et al., "A Thick Volatile Atmosphere on the Ultrahot Super-Earth TOI-561 b", vol. 995, no. 2, p. L39, Dec. 2025, Art. no. L39.  DOI: 10.3847/2041-8213/ae0a4c. arXiv: 2509.17231 [astro-ph.EP].
- 3 J. K. Teske et al., "A Thick Volatile Atmosphere on the Ultrahot Super-Earth TOI-561 b", vol. 995, no. 2, p. L39, Dec. 2025, Art. no. L39.  DOI: 10.3847/2041-8213/ae0a4c. arXiv: 2509.17231 [astro-ph.EP].

Conference Proceedings

- 1 A. McGinty, "Understanding the influence of magma oceans on the observability of atmospheres of rocky exoplanets.", in *EPSC-DPS Joint Meeting 2025*, vol. 2025, Sep. 2025, EPSC-DPS2025-747, EPSC-DPS2025-747.  DOI: 10.5194/epsc-dps2025-747.

- 2 T. G. Meier et al., “Exploring the convection regimes of lava world K2-141b,” in *AASTCS10, Extreme Solar Systems V*, ser. AAS/Division for Extreme Solar Systems Abstracts, vol. 56, Apr. 2024, 609.03, p. 609.03.

Skills

Programming Languages	Python [Advanced], Julia[Beginner], FORTRAN[Beginner], $\LaTeX$ [Adept]
Numerical models	Isca, Unifed Model, SOCRATES, AGNI, petitiRADTRANS
Python libraries	Numpy, xarray, pandas, matplotlib

Outreach

President of Space Exe

- President of the University of Exeter’s astronomy student society. Led star-gazing nights for both students and members of the general public. Give astronomy outreach talks at various events, including the Sidmouth science festival. Organised the return of the Space Exe festival in 2023, an annual student-led outreach event on Streatham campus featuring five guest speakers and a talk on my master’s project.

School talks

- I have given multiple astronomy outreach talks for a range of ages across schools in South Wales. This is a region of the UK where higher education participation remains low, so I am passionate about widening participation.

Miscellaneous Experience

Awards and Achievements

- 2023 **RAS Undergraduate Summer Bursaries**, I was awarded £1000 to undertake a research project with Sebastiaan Krijt on the role of stick organics in planet formation via pebble accretion. I had to turn down this project to pursue my internship at the University of Oxford.
- 2019 **CREST gold Award**, Research into the challenges of detecting gravitational waves - Cardiff School of Physics and Astronomy supervised by Chris North

References

Available on Request